TATA STEEL

Tata Steel Limited
Jamshedpur, Jharkhand 831001
Corporate Identity Number: L27100MH1907PLC000260, Website www.tatasteel.com

IS 10748
CM/L-1488061

TEST CERTIFICATE

TEST CERTIFICATE NO. : 9080007979	DESPATCH DATE: 18.01.2024	SHIP-TO PARTY: 117509	SOLD-TO PARTY: 115806
DO NO. : 9016647426	CUSTOMER CODE: 117509	NARESH STEEL INDUSTRIES PVT. LTD C/O POONYA STEEL	NARESH STEEL INDUSTRIES PVT LTD C-501, NEELKANTH
INVOICE NO. : 2483438104	VEHICLE NO.: MH48AG9078(R)	A-7/2,MIDC,OPP TRANSPOWER,TALOJA	OPP.VIDYA VIHAR RAILWAY STN,
MATERIAL No. : 1108668	PRODUCT: HOT_ROLLED_COILS	TALOJA 410208	MUMBAI 400086

We hereby certify that the above mentioned material is tested in accordance with scheme of testing and inspection contained in BIS certification marks License no. CM/L-1488061 are as indicated against each section (Please refer to IS 10748 GRADE 1(HOT-ROLLED STEEL STRIP FOR WELDED TUBES AND PIPES:2004)* for details of specification requirements). Legal sanction can be imposed on forging. Improper use of product can cause safety issues. To validate test certificate, Scan QR Code.

TDC No: HG01 Q Code:I09497

Grade: IS 10748 Gr 1

Section(TxWxL): 2.000x1250.000x0mm

Chemical Analysis, Specifications and Other Information

	C	Mn	N	P	S	Si	Al	Cu	Cr	Ni	Mo
	%	%	PPM	%	%	%	%	%	%	%	%
Specification Requirements as per TDC No HG01 Min							0.0200				
Max	0.1000	0.5000	120	0.0400	0.0400						
Itm Batch No. Mother Coil Cast No. Tonnage Pcs.											
01 35391171 35391171 V94364 22.550 1	0.0350	0.1660	30	0.0180	0.0080	0.0050	0.0690	0.0070	0.0210	0.0190	0.0001

TDC No: HG01

Mechanical Analysis, Specifications and Other Information

Section (TxWxL): 2.000x1250.000x0mm

	BEND	EL	YS	UTS
		%	MPa	MPa
Speciifcation Requirements as per TDC No. HG01 Min		30	170	290
Max				
Itm Batch No. Mother Coil Cast No. Tonnage				
01 35391171 35391171 V94364 22.550 1	OK	38	275	349

1Y1YBK 220

Route-BOF>LF>CC>HSM
Test Certificate as per EN10204 TYPE 3.1

1 MPa = 1 N/mm2
YS = Yield Strength
UTS = Ultimate Tensile Strength

El= % elongation on standard gauge length
TxWxL = Thickness x Width X Length
CE (Carbon Equivalent) = (C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15)

*For further details please visit www.bis.gov.in

Killing type:Steel is fully Aluminium killed
Material is ROHS Compliant
Tensile Sample testing transverse to rolling direction

QC Verified

Material clearance date

04.01.2024
This TC is electronically generated

Date:2023.01.1815:53:27IST

AM/NS INDIA

ArcelorMittal Nippon Steel India Limited PRECOATED FACILITY

Works: Gat No.740, Sahajpur, Kadar Road, Pune, 412201

CIN:U27100GJ1976PLC013783

MILL TEST CERTIFICATE

FOR GALVANIZED STEEL STRIPS AND SHEETS(PLAIN & CORRUGATED)

Test Certificate No.	AMNSIECO/2024/307094/0246/350/00
Sale Order No./Item No.	2120487696 / 40
Equivalent Specification	IS_277_GP250
Test Certificate Date	07.03.2024
Sales Order Date	05.03.2024
Material Description	GALVANIZED STEEL STRIP
Invoice No.	MH202310000025768
Vehicle No.	MH46BF6183

To,
K. KRISHNA STEEL INDUSTRIES
A-103, 1st FLOOR, RATNA RAJUL CHS L
400067
MUMBAI 13
India.

WE CERTIFY THAT THE MATERIAL DESCRIBED BELOW FULLY CONFORMS TO ABOVE SPECIFICATION CHEMICAL COMPOSITION AND MECHENICAL PROPERTIES OF THE PRODUCT, AS TESTED IN ACCORDANCE WITH THE SCHEME OF TESTING AND INSPECTION CONTAINED IN THE BIS CERTIFICATION MARKS LICENCE NO. CM/L-3784580 ARE AS INDICATED BELOW AGAINST EACH ORDER NO. IFTC.

Heat Analysis (%) of AN24100645

Heat No	C	Mn	S	P	Si	Al
Min-->	N.S	N.S	N.S	N.S	N.S	0.020
Max-->	0.080	0.500	0.0250	0.025	0.040	0.080
AN24100645	0.057	0.215	0.0047	0.013	0.005	0.045

Mechanical/Physical Properties

Item No	Batch ID	Thk	Wid	Len	Qty Net	HARD	IMPT	BEND	ST	CFT	ZCOAT
Min-->						N.S	N.S	N.S	N.S	N.S	N.S
Max-->						N.S	N.S	N.S	N.S	N.S	N.S
40	2501SB003P	0.900	1250	652	5.560	58	OK	OK	C	N	60
Total					5.560						

Note :
Due to ageing, The coating surface can vary & change to a dark appearance by oxidation. Reduction in formability, cracking of the surface, fluting can occur during processing. Therefore the user should process the material as quick as possible

QC Verified

Legends: Bend: Bend Test Table no 5 (Clause no 8.3), CFT: Coating Finish Type, IMPT: Impact Test (100 Kg-Cm), ST: Surface Treatment, ZCOAT: Zinc Coating in GSM, N.S: Not Specified

Units of measurement: Thk/Wid(MM), Len(MTR), Qty Net(Quantity Net)(MT), HARD(HRB), ZCOAT(GSM)

MANAGER (QUALITY ASSURANCE)
FOR ArcelorMittal Nippon Steel India Limited PRECOATED FACILITY

Scanned with ACE Scanner

ArcelorMittal Nippon Steel India Limited

HAZIRA - 394270, DIST. SURAT, GUJARAT, INDIA

CIN:U27100GJ1985PLC009767

MILL TEST CERTIFICATE

FOR For GALVANIZED STEEL SHEETS/STRIPS

CM/L 7617379

Test Certificate No.	AMNS/TCG/20220722164922705930	Test Certificate Date	22.01.2024	To, ABHAY ISPAT INDIA PVT LTD 3RD FLOOR, 308, PLATINUM, 400077 MUMBAI 13 India.
Sale Order No.	2120422893 / 20	Sales Order Date	20.01.2024	
Equivalent Specification	IS 277 2018 GP	Product Type	HOT-DIP GALVANIZED COIL	
Specification	IS 277 2018 GP			
Invoice No.	GJ20221000057710			
Vehicle No.	MH04GF8403			

WE CERTIFY THAT THE MATERIAL DESCRIBED BELOW FULLY CONFORMS TO ABOVE SPECIFICATION. CHEMICAL COMPOSITION AND MECHANICAL PROPERTIES OF THE PRODUCT, AS TESTED IN ACCORDANCE WITH THE SCHEME OF TESTING AND INSPECTION CONTAINED IN THE BIS CERTIFICATION MARKS LICENCE NO. CM/L 7617379 ARE AS INDICATED BELOW AGAINST EACH ORDER NO. ETC.

Heat Analysis (%) of 22402219

Heat No	C	MN	S	P	SI
Min-->	-	-	-	-	-
Max-->	0.100	0.500	0.0350	0.030	0.040
22402219	0.051	0.221	0.0024	0.009	0.017

Mechanical Properties for Batches of 22402219

Item No	Batch ID	Thk	Wid	Qty Grs	Qty Net	BEND	HARD	COAT
Min-->						-	-	80.00
Max-->						-	65.00	-
20	2U1970200G	1.200	1250	5.305	5.275	OK	64	93
Total				5.305	5.275			

HS-1361

Note 1) Test Certificate conforms to EN 10204-3.1

QC Verified

Legends: Bend:0, COAT: Coating in GSM, HARD: Hardness in HRB

Units of measurement: Thk/Wid/Len(MM), Qty Grs/Net(Quantity Gross/Net)(MT), COAT(GSM)

MANAGER (QUALITY ASSURA)
FOR ArcelorMittal Nippon Steel India Limited
AMNS/L/QAC/0001/F

*This document is electronically generated and hence does not require any signature

	TEST CERTIFICATE FOR STEEL TUBES	Format No.: QA-F-18 Rev.no. :00 Date :10.12.2023

APL APOLLO TUBES LTD. UNIT-III

Works-Vill-Kudavali Plot No-M-1, Murbad, 421401, Thane (MS)

TEST CERTIFICATE FOR STEEL TUBES

IS: 1161

CM/L-2808361

TEST CERTIFICATE NO. : IS 1161
DATE : 05-01-23
INVOICE NO. : 2204234569
VEHICLE NO. : MH12KQ2728
INVOICE DATE : 25-03-23

M/s H VIKAS PIPE & STEEL LLP Pune

We certify that the material described below fully conforms to IS 1161. Chemical composition and mechanical properties of the product, as tested in accordance with the Scheme of Testing and Inspection contained in the BIS Certification Marks License No. CM/L-2808361 are as indicated below against each order no. etc.

(PLEASE REFER TO IS 1161 FOR DETAILS OF SPECIFICATION REQUIREMENTS)

TEST RESULT

Size	Grade/ Type	Cast no./ Lot No	Total Pcs	Actual Wt (MT)	CHEMICAL ANALYSIS (% Max)				MECHANICAL PROPERTIES (Min)					GALVANIZING (Min)	
					C	Mn	P	S	Tensile Strength (Mpa)	Yeild Strength	%El	Drift Expansion	Bend test/ Flattening Test	Weight (g/m2)	Uniformity
Requirements					N/A	N/A	0.04	0.04	360	210	20	12.5 % at 45°	-	400	5 Dip (1Dip/ Minute)
OD 165X5.00 MM	WT 160 / BLK	**222101154 / 4222414A**	455	9.6	0.037	0.22	0.013	0.013	376	330	34	OK	OK	-	-
OD 140X3.50 MM	WT 160 / BLK	**222101154 / 4222414A**	455	8.5	0.037	0.22	0.013	0.013	374	330	34	OK	OK	-	-

It is certify that each steel tube is supplied conforms to standard dimensions and mass tolerances.

% El : Elongation percentage

QC Verified

AUTHORISED SIGNATORY

APL APOLLO TUBES LTD. MURBAD

PREPARED BY

FOR APL APOLLO TUBES LTD.

	TEST CERTIFICATE FOR STEEL TUBES	Format No.: QA-F-18 Rev.no. : 00 Date :15.12.2023

APL APOLLO TUBES LTD. UNIT-III

Works-Vill-Kudavali Plot No-M-1, Murbad, 421401, Thane (MS)

TEST CERTIFICATE FOR STEEL TUBES

IS: 1161

CM/L-2808361

TEST CERTIFICATE NO. : IS 1161
DATE : 15-01-23
INVOICE NO. : 220423480
VEHICLE NO. : MH12HJ4850
INVOICE DATE : 10-04-23

M/s H VIKAS PIPE & STEEL LLP Pune

We certify that the material described below fully conforms to IS 1161. Chemical composition and mechanical properties of the product, as tested in accordance with the Scheme of Testing and Inspection contained in the BIS Certification Marks License No. CM/L-2808361 are as indicated below against each order no. etc.

(PLEASE REFER TO IS 1161 FOR DETAILS OF SPECIFICATION REQUIREMENTS)

TEST RESULT

Size	Grade/ Type	Cast no./ Lot No	Total Pcs	Actual Wt (MT)	CHEMICAL ANALYSIS (% Max)				MECHANICAL PROPERTIES (Min)					GALVANIZING (Min)	
					C	Mn	P	S	Tensile Strength (Mpa)	Yeild Strength	%El	Drift Expansion	Bend test/ Flattening Test	Weight (g/m2)	Uniformity
Requirements					N/A	N/A	0.04	0.04	360	210	20	12.5 % at 45°	-	400	5 Dip (1Dip/ Minute)
OD 114X3.00 MM	WT 160 / BLK	222101154 / 4222414A	455	7.6	0.037	0.22	0.013	0.013	374	330	34	OK	OK	-	-

It is certify that each steel tube is supplied conforms to standard dimensions and mass tolerances.

% El : Elongation percentage

QC Verified

AUTHORISED SIGNATORY

APL APOLLO TUBES LTD. MURBAD

PREPARED BY

FOR APL APOLLO TUBES LTD.

STEEL AUTHORITY OF INDIA LIMITED
ROURKELA STEEL PLANT
ROURKELA - 769011 (ODISHA)
RESEARCH AND CONTROL LABORATORY

IS 2062

Test Certificate For: HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL STEEL

Test Certificate No.: 436945 | Dated: 26.06.2022 | Product: HR Plate

To : LITTLESTAR PROJECTS PVT LTD C-12, SETH DEOKARAN DASS ROURKELA 769012

It is cerified that the materials described below fullyconforms to IS 2062 - 2011. The chemical composition and Mechanical properties of the product as tested in accordance with the scheme of Testing and Inspection contained in BIS Certification marks License No. CM / L 0000053223 are as indicated against each order. Thematerial supplied conforms to standard Rolling and weight tolerances. (PLEASE REFER TO IS 2062 - 2011 FOR DETAILS OF SPECIFICATION REQUIREMENTS)

Ladle Analysis

Cast No	C%	Mn%	P%	S%	Si%	Al%	CE
2009864	0.1400	0.7900	0.0290	0.0190	0.1080	0.0290	0.2680
2009889	0.1400	0.8700	0.0275	0.0150	0.1570	0.0170	0.2780

Sl. No.	MP / SO No.	Cast / Batch No.	Size(mm) (TxWxL)	Weight(T)	Grade / Mode Of Killing	YS (MPa)	UTS (MPa)	El (%)	Bend Test
1	1400462204	2009884 / N9761801	1250X2500X5 MM	26.45	IS 2062 E250BR / ALSIK	294	449	34	OK
2	1400462204	2009884 / N9761801	1250X2500X3 MM	30.20	IS 2062 E250BR / ALSIK	291	444	32	OK

Total Pieces : 20 **Total Weight(Tons) :** 56.65

Remarks : PROCESS ROUTE: BOF-LHF-CC-HSM

QC Verified

K PATWARI

DGM , R&C Lab
Rourkela Steel Plant

Dispatch Advice No.: 811307979 Dated: 26.06.2022 Truck No: OD14P1023